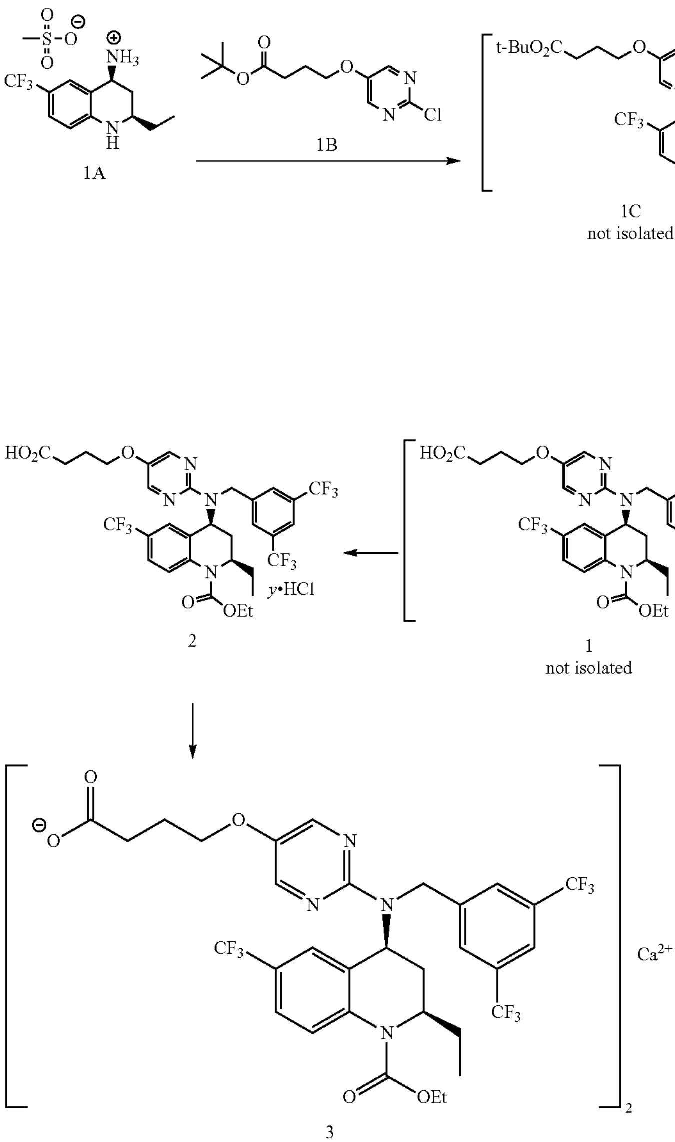
Scheme 1



[0493] With reference to Scheme 1, amorphous obicetrapib hemicalcium (compound 3) was prepared in six chemical steps and three isolations from the mesylate salt of (2R,4S)-4-amino-2-ethyl-6-trifluoromethyl-3,4-dihydro-2H-quinoline (compound 1A), t-butyl-4-(2-chloropyrimidin-5-yloxy)-butyrate (compound 1B), and 3,5-bis(trifluoromethyl)benzyl bromide (compound 1E).

[0494] Compound 1A was coupled with compound 1B through a palladium-catalyzed reaction to produce a solution of (2R,4S)-4-[5-(3-t-butoxycarbonylpropoxy)pyrimidin-2-yl)]amino-2-ethyl-6-trifluoromethyl-3,4-dihydro-2H-quinoline (compound 1C), which was not isolated but directly reacted with excess ethyl chloroformate in the presence of pyridine to produce (2R,4S)-4-[5-(3-t-butoxycarbonylpropoxy)pyrimidin-2-yl)]amino-2-ethyl-6-trifluoromethyl-3,4-dihydro-2H-quinoline-1-carboxylic acid ethyl ester, which was isolated as a crystalline mesylate salt (Compound 1D). The crystalline mesylate salt, Compound 1D was alkylated with 3,5 bis(trifluoromethyl)benzyl bromide (compound 1E) under strongly basic conditions to produce a solution of (2R,4S)-4-{[3,5-bis(trifluoromethyl)benzyl]-[5-(3-t-butoxycarbonylpropoxy)pyrimidin-2-yl]amino}-2-ethyl-6-trifluoromethyl-3,4-dihydro-2H-quinoline-1-carboxylic acid ethyl ester (compound 1F) in toluene. Compound IF was then subjected to an acidic cleavage of the tert-butyl ester to produce a solution of (2R,4S)-4-{[3,5-bis(trifluoromethyl)benzyl]-[5-(3-carboxypropoxy)pyrimidin-2-yl]amino}-2-ethyl-6-trifluoromethyl-3,4-dihydro-2H-quinoline-1-carboxylic acid ethyl ester (compound 1). Compound 1 was then converted to compound 2, which is a solvate of (2R,4S)-4-{[3,5-bis(trifluoromethyl)benzyl]-[5-(3-carboxypropoxy)pyrimidin-2-yl]amino}-2-ethyl-6-trifluoromethyl-3,4-dihydro-2H-quinoline-1-carboxylic acid